1-Number of Zeros in a Given Array

Started on	Monday, 31 August 2026, 4:29 PM
State	Finished
Completed on	Saturday, 5 September 2026, 8:16 AM
Time taken	4 days 15 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

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```c
#include <stdio.h>

int firstZero(int a[], int low, int high)
{
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (a[mid] == 0)
    {
        if (mid == 0 || a[mid - 1] == 1)
            return mid;

        return firstZero(a, low, mid - 1);
    }

    return firstZero(a, mid + 1, high);
}

int main()
{
    int m;
    scanf("%d", &m);

    int a[m];

    for (int i = 0; i < m; i++)
        scanf("%d", &a[i]);

    int index = firstZero(a, 0, m - 1);

    if (index == -1)
        printf("0\n");
    else
        printf("%d\n", m - index);

    return 0;
}
```

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	Input	Expected	Got	
☐	5 1 1 1 0 0	2	2	☐
☐	10 1 1 1 1 1 1 1 1 1 1	0	0	☐
☐	8 0 0 0 0 0 0 0 0	8	8	☐

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	0 0 0			
☐	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	☐

Passed all tests! ☐

Correct

Marks for this submission: 1.00/1.00.

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2-Majority Element

Started on	Saturday, 5 September 2026, 8:17 AM
State	Finished
Completed on	Saturday, 5 September 2026, 8:17 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct | Mark 1.00 out of 1.00 | Flag question

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

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Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

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```c
int main()
{
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    int candidate = 0;
    int count = 0;

    for (int i = 0; i < n; i++)
    {
        if (count == 0)
            candidate = nums[i];

        if (nums[i] == candidate)
            count++;
        else
            count--;
    }

    printf("%d\n", candidate);

    return 0;
}
```

	Input	Expected	Got	
	3 3 2 3	3	3	

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3-Finding Floor Value

Started on	Saturday, 5 September 2026, 8:18 AM
State	Finished
Completed on	Saturday, 5 September 2026, 8:22 AM
Time taken	4 mins 23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int a[], int low, int high, int x)
{
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] > x)
        return findFloor(a, low, mid - 1, x);

    int result = findFloor(a, mid + 1, high, x);

    if (result == -1)
        return a[mid];

    return result;
}

int main()
{
    int n, x;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d\n", findFloor(a, 0, n - 1, x));

    return 0;
}
```

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	Input	Expected	Got	
	6 1 2 8 10 12 19 5	2	2	
	5 10 22 85 108 129 100	85	85	
	7 3 5 7 9 11 13 15 10	9	9	

Passed all tests!

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4-Two Elements sum to x

Started on	Saturday, 5 September 2026, 8:23 AM
State	Finished
Completed on	Saturday, 5 September 2026, 8:26 AM
Time taken	2 mins 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

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```c
#include <stdio.h>
int findPair(int a[], int left, int right, int x, int *e1, int *e2)
{
    if (left >= right)
        return 0;
    int sum = a[left] + a[right];
    if (sum == x)
    {
        *e1 = a[left];
        *e2 = a[right];
        return 1;
    }
    if (sum > x)
        return findPair(a, left, right - 1, x, e1, e2);
    return findPair(a, left + 1, right, x, e1, e2);
}
int main()
{
    int n, x;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    int e1, e2;

    if (findPair(a, 0, n - 1, x, &e1, &e2))
    {
        printf("%d\n", e1);
        printf("%d\n", e2);
    }
    else
    {
        printf("No\n");
    }

    return 0;
}
```

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	Input	Expected	Got	
	4 2 4 8 10 14	4 10	4 10	
	5 2 4 6 8 10 100	No	No	

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Correct

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Started on	Saturday, 5 September 2026, 8:26 AM
State	Finished
Completed on	Saturday, 5 September 2026, 8:33 AM
Time taken	7 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

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Answer:

```c
#include <stdio.h>
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int a[], int low, int high)
{
    int pivot = a[high];
    int i = low - 1;
 for (int j = low; j < high; j++)
    {
        if (a[j] < pivot)
        {
            i++;
            swap(&a[i], &a[j]);
        }
    }
swap(&a[i + 1], &a[high]);
    return i + 1;
}
void quickSort(int a[], int low, int high)
{
    if (low < high)
    {
        int p = partition(a, low, high);

        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}
int main()
{
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
```

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```
        scanf("%d", &a[i]);
    quickSort(a, 0, n - 1);

    for (int i = 0; i < n; i++)
    {
        printf("%d", a[i]);

        if (i < n - 1)
            printf(" ");
    }

    return 0;
}
```

	Input	Expected	Got	
	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	
	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	
	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	

Passed all tests!

Correct

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